



Dr. Ingo Frommholz

Professor of Applied Data Science

PhD (Dr. rer. nat.)

Diplom-Informatiker

FBCS, FHEA

At a Glance

- Full Professor and Head of School of Applied Data Science, Modul University Vienna, Austria; Adjunct Professor (Lehrbeauftragter) for Data Engineering at Bern University of Applied Sciences, Switzerland
- Interested in user-oriented Information Retrieval, AI, NLP and Data Engineering with application for example in academic search and recommendation, digital libraries, social sciences, e-commerce, and digital humanities
- Chair of the BCS Information Retrieval Specialist Group (BCS-IRSG)
- Founding Head of the Data Science and AI Research group (DAIREL), University of Wolverhampton, UK (2023–2025)
- Steering committee member, ACM CIKM (Core-A-ranked conference) and ACM SIGIR ICTIR conference series
- Funding (PI): EU Horizon Europe MSCA Staff Exchange project OMINO on Information Overload (2023–2026, UKRI guarantee funding, total EUR 1,550,200, own part EUR 253,000); EU H2020 ETN QUARTZ on quantum mechanics and information retrieval (2016–2020, total EUR 3,461,495, own part EUR 546,758); internally funded RIGHT project on deep learning for digital humanities (since 2023, GBP 612,621); plus further smaller research and knowledge transfer projects.
- Published more than 100 refereed and unrefereed articles, preprints, reports, proceedings, books and book sections
- Senior Managing Editor of the International Journal on Digital Libraries (Springer)
- General Chair, ACM CIKM 2023, Birmingham (Core-A-ranked conference with 750 delegates) and ELLIS ESSIR 2025; Programme Chair, ACM/IEEE JCDL 2025; Short Paper Programme Chair, ECIR 2027 (Southampton, UK); Organiser of several international workshops and schools (e.g. BIR, BIRNDL, BIRDS, BCS Search Solutions, FDIA, ASIRF)
- Chair and senior programme committee member for several world-leading conferences and journals in Information Retrieval, AI and Digital Libraries
- Course and programme co-ordinator for Computer Science, Data Science and Artificial Intelligence undergraduate and postgraduate programmes; module co-ordinator and lecturer in several Computer Science topics at undergraduate and postgraduate level in the UK, Austria, Germany, Switzerland, Oman and Jordan
- Supervised 12 PhD students and 2 MSc-by-research students to successful completion; currently supervising 5 PhD students (1 as first supervisor, 4 as external supervisor); examiner for 15 PhD candidates in the UK, Denmark, France, Finland, The Netherlands, Italy, Germany and Austria
- External examiner: University of Sheffield (since 2023); University of Strathclyde (2020–2024)
- PhD in Computer Science (Dr. rer. nat., summa cum laude), University of Duisburg-Essen, Germany
- Fellow of the BCS (FBCS) and Advance HE (FHEA); member of ACM and ACM SIGIR

Education

- 2008 **PhD (Dr. rer. nat.) in Computer Science**, *University of Duisburg-Essen*, Duisburg, Germany
 2001 **Diplom-Informatiker (MSc Computer Science equivalent)**, *University of Dortmund*, Germany

Professional Experience

- 03/2025–
Present **Professor of Applied Data Science and Head of School**, *School of Applied Data Science, Modul University*, Vienna, Austria
- 02/2023–
Present **Adjunct Professor (Lehrbeauftragter) for Data Engineering**, *School of Engineering and Computer Science, Bern University of Applied Sciences*, Biel/Bienne, Switzerland
- 2023–2026 **Visiting researcher (OMINO secondments)**, *Nanyang Technological University (Singapore), University of New Hampshire (USA), GESIS – Leibniz Institute for the Social Sciences (Germany) and further project partners*
- 12/2020–
02/2025 **Reader (Associate Professor equivalent) in Data Science**, *School of Engineering, Computing and Mathematical Sciences, University of Wolverhampton*, UK
- 02/2011–
11/2020 **Lecturer/Senior Lecturer**, *School of Computer Science and Technology, University of Bedfordshire*, Luton, UK
- 2012–2018 **Search and Data Science Consultant**, *Self Employed*, London, UK
- 11/2008–
01/2011 **Postdoctoral Research Associate**, *Information Retrieval Group, School of Computing Science, University of Glasgow*, Glasgow, UK
- 10/2004–
10/2008 **Research Assistant**, *Information Systems Group, Faculty of Engineering Sciences, University of Duisburg-Essen*, Duisburg, Germany
- 03/2006–
04/2006 **Visiting researcher**, *Information Management Systems Research Group, University of Padua*, Italy
- 03/2001–
09/2004 **Research Assistant**, *Fraunhofer Integrated Publication and Information Systems Institute (IPSI)*, Darmstadt, Germany
- 11/1994–
02/2001 **Software Developer**, *Several companies: adesso GmbH, o.tel.o GmbH, Lion Ges. für Systementwicklung mbH*, Bochum, Düsseldorf, Dortmund, Germany

Selected Grants and Funding

Research Grants (as PI / Lead)

- 2027–2030 **TrustLabel.AI**, *Modul University Vienna*, Austria **EUR 822,938**
 FFG-funded project (MU part EUR 245,180).
- 2023–2026 **OMINO**, *Univ. of Wolverhampton & Modul University Vienna*, UK/Austria **EUR 1,550,200**
 EU Horizon Europe MSCA Staff Exchange project (guarantee funded by UKRI/EPSC); UoW part GBP 218,036 (EUR 253,000).
- 2023–2024 **RIGHT**, *University of Wolverhampton*, UK **GBP 612,621**
 Internally funded project.
- 2023–2024 **LASER**, *University of Wolverhampton*, UK **GBP 39,000**
 Responsible Digital Humanities Lab. Internally funded PhD project.
- 2016–2020 **QUARTZ**, *University of Bedfordshire*, Luton, UK **EUR 3,461,495**
 EU H2020 Marie Skłodowska-Curie ITN European Training Network; own part GBP 473,387 (EUR 546,576).
- 2012–2013 **Rising Star**, *University of Bedfordshire*, Luton, UK **GBP 6,000**
 Internal Research Investment Rising Star Funding.
- 2010–2011 **QONTEXT**, *University of Glasgow*, Glasgow, UK
 EU Marie Curie FP7-IRSES project.

2004–2006	CLASSIX , <i>University of Duisburg-Essen</i> , Duisburg, Germany German Research Foundation (DFG) project. Knowledge Transfer and Industry Grants (as PI / Lead)	
2016–2019	Drax Technology Ltd , <i>University of Bedfordshire</i> , Luton, UK Innovate UK Knowledge Transfer Partnership with Drax Technology.	GBP 134,752
2012–2018	NotActivelyLooking.com Private consultancy project (in collaboration with industry partners).	ca. GBP 20,000
2016–2017	Wardown Park Talking Objects , <i>University of Bedfordshire</i> , Luton, UK With Wardown Park Museum, Luton (Luton Cultural Services Trust).	GBP 18,971
2015	EduVocation , <i>University of Bedfordshire</i> , Luton, UK Teacher recruitment site and information system based on linked open data (UK Innovation Voucher).	GBP 5,000
2013–2015	Luton Culture World War I project , <i>University of Bedfordshire</i> , Luton, UK With Luton Museum (UK Lottery Fund).	GBP 16,000
2006	KIIB , <i>University of Duisburg-Essen</i> , Duisburg, Germany EU-EUREGIO project in collaboration with d.velop AG, Gescher, Germany.	

Further Research Projects: EDISON (University of Bedfordshire): EU H2020 Coordination and Support Action to accelerate the creation of the Data Science profession; RENAISSANCE (University of Glasgow): EPSRC project on retrieval models based on the quantum theory framework; LACOSTIR (University of Duisburg-Essen): DFG project on interactive retrieval; DELOS (University of Duisburg-Essen): EU-FP6 Network of Excellence on Digital Libraries; BRICKS (Fraunhofer IPSI): EU-FP6 integrated project on distributed resources for integrated cultural knowledge services; VIP-ADVISOR (Fraunhofer IPSI): EU-FP5 project (IST-2001-32440) on a virtual advisor for personal insurance and finance risk management; COLLATE (Fraunhofer IPSI): EU-FP5 project (IST-1999-20882) on the development of a collaboratory for annotation, indexing and retrieval of digitized historical archive material; SCHOLNET (Fraunhofer IPSI): EU-FP5 project (IST-1999-20664) on a digital library infrastructure for supporting communication and collaboration among networked scholarly communities.

Further Knowledge Transfer Projects: SOLVD (University of Wolverhampton): EU European Regional Development Fund to support SMEs in the adoption of digital technologies; CRASH (University of Bedfordshire): Cyber Risk Assessment of Stalking and Harassment.

Selected Publications

Published more than 100 refereed and unrefereed articles, preprints, reports, proceedings, books and book sections. The below selection reflects publications that represent my research interests.

- [1] Przemysław Kazienko, Wiktoria Mieszczenko-Kowszewicz, Jan Kocoń, Beata Bajcar, Julian Sienkiewicz, Janusz Hołyst, Philipp Mayr, Ingo Frommholz, and Erik Cambria. Artificial Intelligence Overload: A Multilevel Taxonomy and the Path Forward. *IEEE Intelligent Systems*, 41(3):5–12, May 2026.
- [2] Kevin Schott, Kanishka Silva, Ingo Frommholz, Philipp Mayr, Dagmar Kern, and Daniel Hienert. AI Overviews in Academic Search: Evaluating AI-generated Summaries of Search Results in a Domain-specific Search Engine. *Proceedings of the Association for Information Science and Technology*, 63(1), 2026.
- [3] Damith Premasiri, Tharindu Ranasinghe, Ruslan Mitkov, Mo El-Haj, and Ingo Frommholz. Survey on legal information extraction: current status and open challenges. *Knowledge and Information Systems*, October 2025.
- [4] Amit Kumar Jaiswal, Haiming Liu, and Ingo Frommholz. Multimodal RAG Enhanced Visual Description. In *Proceedings of the ACM Conference on Information and Knowledge Management (CIKM 2025)*, Seoul, Korea, August 2025. ACM. arXiv:2508.09170 [cs].
- [5] Kanishka Silva, Ingo Frommholz, Burcu Can, Fred Blain, Raheem Sarwar, and Laura Ugolini. Forged-GAN-BERT: Authorship Attribution for LLM-Generated Forged Novels. In Neele Falk, Sara Papi, and Mike Zhang, editors, *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics*

(EACL 2024): *Student Research Workshop*, pages 325–337, St. Julian's, Malta, March 2024. Association for Computational Linguistics.

- [6] Janusz A. Hołyst, Philipp Mayr, Michael Thelwall, Ingo Frommholz, Shlomo Havlin, Alon Sela, Yoed N. Kenett, Denis Helic, Aljoša Rehar, Sebastijan R. Maček, Przemysław Kazienko, Tomasz Kajdanowicz, Przemysław Biecek, Bolesław K. Szymanski, and Julian Sienkiewicz. Protect our environment from information overload. *Nature Human Behaviour*, 8:402–403, February 2024.
- [7] Julius Odede and Ingo Frommholz. JayBot – Aiding University Students and Admission with an LLM-based Chatbot. In *Proceedings ACM SIGIR Conference on Human Information Interaction And Retrieval (CHIIR 2024)*, pages 391–395. ACM, 2024.
- [8] Karen Chessum, Haiming Liu, and Ingo Frommholz. An Extended Study of Search User Interface Design Focused on Hofstede's Cultural Dimensions. In Andreas Holzinger, Hugo Plácido Da Silva, Jean Vanderdonckt, and Larry Constantine, editors, *Computer-Human Interaction Research and Applications*, volume 1882, pages 130–152, Cham, 2023. Springer Nature Switzerland. Series Title: Communications in Computer and Information Science.
- [9] Kanishka Silva and Ingo Frommholz. What if ChatGPT wrote the Abstract? - Explainable Multi-Authorship Attribution with a Data Augmentation Strategy. In *Proceedings of the IACT - The 1st International Workshop on Implicit Author Characterization from Texts for Search and Retrieval (IACT@SIGIR 2023)*, pages 38–48, Taipei, Taiwan, 2023. CEUR-WS.org.
- [10] Nicholas Collis and Ingo Frommholz. Crowdsourced Linked Data Question Answering with AQUACOLD. In *Proceedings of the ACM/IEEE Joint Conference on Digital Libraries (JCDL 2021)*. ACM, 2021.
- [11] Amit Kumar Jaiswal, Haiming Liu, and Ingo Frommholz. Semantic Hilbert Space for Interactive Image Retrieval. In *ICTIR 2021 - Proceedings of the 2021 ACM SIGIR International Conference on Theory of Information Retrieval*, pages 307–315, 2021.
- [12] Amit Kumar Jaiswal, Haiming Liu, and Ingo Frommholz. Utilising Information Foraging Theory for User Interaction with Image Query Auto-Completion. In *Proceedings European Conference on Information Retrieval (ECIR 2020)*. Springer, 2020.
- [13] Ingo Frommholz, Haider Al-Khateeb, Martin Potthast, Zinnar Ghasem, Mitul Shukla, and Emma Short. On Textual Analysis and Machine Learning for Cyberstalking Detection. *Datenbank-Spektrum*, 16(2):127–135, 2016.
- [14] Ingo Frommholz and Thomas Roelleke. Scalable DB+IR Technology: Processing Probabilistic Datalog with HySpirit. *Datenbank-Spektrum*, 16(1):39–48, 2016.
- [15] Muhammad Kamran Abbasi and Ingo Frommholz. Cluster-based Polyrepresentation as Science Modelling Approach for Information Retrieval. *Scientometrics*, 102(3):2301–2322, 2015.
- [16] Ingo Frommholz, Birger Larsen, Benjamin Piwowarski, Mounia Lalmas, Peter Ingwersen, and Keith Van Rijsbergen. Supporting Polyrepresentation in a Quantum-inspired Geometrical Retrieval Framework. In Paul Clough, Colum Foley, Cathal Gurrin, Gareth J. F. Jones, and Wessel Kraaij, editors, *IliX '10: Proceedings of the third symposium on Information Interaction in Context (IliX 2010)*, pages 115–124. ACM, 2010.
- [17] Benjamin Piwowarski, Ingo Frommholz, Mounia Lalmas, and Keith van Rijsbergen. What can Quantum Theory bring to Information Retrieval? In *CIKM '10: Proceedings of the 19th ACM International Conference on Information and Knowledge Management*, pages 59–68, 2010.
- [18] Ingo Frommholz and Norbert Fuhr. Probabilistic, Object-oriented Logics for Annotation-based Retrieval in Digital Libraries. In Michael Nelson, Cathy Marshall, and Gary Marchionini, editors, *Proc. of the 6th ACM/IEEE Joint Conference on Digital Libraries (JCDL 2006)*, pages 55–64, New York, 2006. ACM.
- [19] Maristella Agosti, Nicola Ferro, Ingo Frommholz, and Ulrich Thiel. Annotations in Digital Libraries and Collaboratories – Facets, Models and Usage. In Rachel Heery and Liz Lyon, editors, *Research and Advanced Technology for Digital Libraries. Proc. European Conference on Digital Libraries (ECDL 2004)*, pages 244–255, Heidelberg et al., 2004. Springer. Series Title: Lecture Notes in Computer Science.
- [20] Ulrich Thiel, Holger Brocks, Ingo Frommholz, Andrea Dirsch-Weigand, Jürgen Keiper, Adelheit Stein, and Erich Neuhold. COLLATE - A Collaboratory Supporting Research on Historic European Films. *International Journal on Digital Libraries (IJDL)*, 4(1):8–12, 2004. Series Number: 1.